

MATH 3312, SPRING 2026: LECTURE 6, CYCLICITY OF $k^\times$ FOR k FINITE

Theorem 1. Suppose that k is a finite field; then $k^\times$ is cyclic.

Proof. The proof will use the following criterion for cyclicity:

Claim. The following are equivalent:

- (1) $k^\times$ is cyclic;
- (2) $k^\times$ has one invariant factor;
- (3) For every $d \geq 1$, the set $\{\alpha \in k : \alpha^d = 1\}$ has exactly d elements;
- (4) For every $d \geq 1$, the set $\{\alpha \in k : \alpha^d = 1\}$ has *at most* d elements.

This follows essentially from material from last semester, but let us quickly recall the details. By the fundamental theorem of finitely generated abelian groups, we have

$$k^\times \simeq \mathbb{Z}/d_1\mathbb{Z} \times \cdots \times \mathbb{Z}/d_r\mathbb{Z}$$

where $2 \leq d_1 \mid \cdots \mid d_r$ are the invariant factors. Therefore, $k^\times$ is cyclic exactly when $r = 1$. Moreover, if $r \geq 2$, then each factor would be killed by d_r , and so the set of elements killed by d_r would have *strictly more* than d_r elements. The contrapositive is that *if* there are no more than d_r elements killed by d_r then we must have a single invariant factor. This shows that assertion (4) of the claim implies (1).

Now, we will verify that $k^\times$ satisfies condition (4) of the claim. This follows from Problem 8 on Homework 1 applied to the degree d polynomial $x^d - 1 \in k[x]$. □

Proposition: $K^\times$ is cyclic.

Pl.: $K^\times$ is finite & abelian

$$\Rightarrow \underline{K^\times} \cong \underbrace{\left[(\mathbb{Z}/d_1\mathbb{Z})^\times \cdots \right]}_{\substack{\text{FT of} \\ \text{finite abelian} \\ \text{groups}}} \underbrace{(\mathbb{Z}/d_r\mathbb{Z})^\times}_{\substack{d_1, d_2, \dots, d_r \\ \text{(invariant} \\ \text{factors)}}}$$

TFAE:

- (1) $K^\times$ is cyclic
- (2) $r=1$ (there is one invariant factor)
- (3) $\nexists d \mid |K^\times|$

$$\textcircled{3} \quad \left| \{ \alpha \in k^\times : \alpha^d = 1 \} \right| = d$$

$$\forall d \mid |k^\times|, \left| \{ \alpha \in k^\times : \alpha^d = 1 \} \right| \leq d$$

$$\{ \alpha \in k^\times : \alpha^d = 1 \}$$

$$= \left\{ \alpha \in k : \alpha \text{ is a } \underline{\text{zero}} \text{ for } x^d - 1 \right\}$$

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$$

there are 8 elements
(not 4) killed by 4.

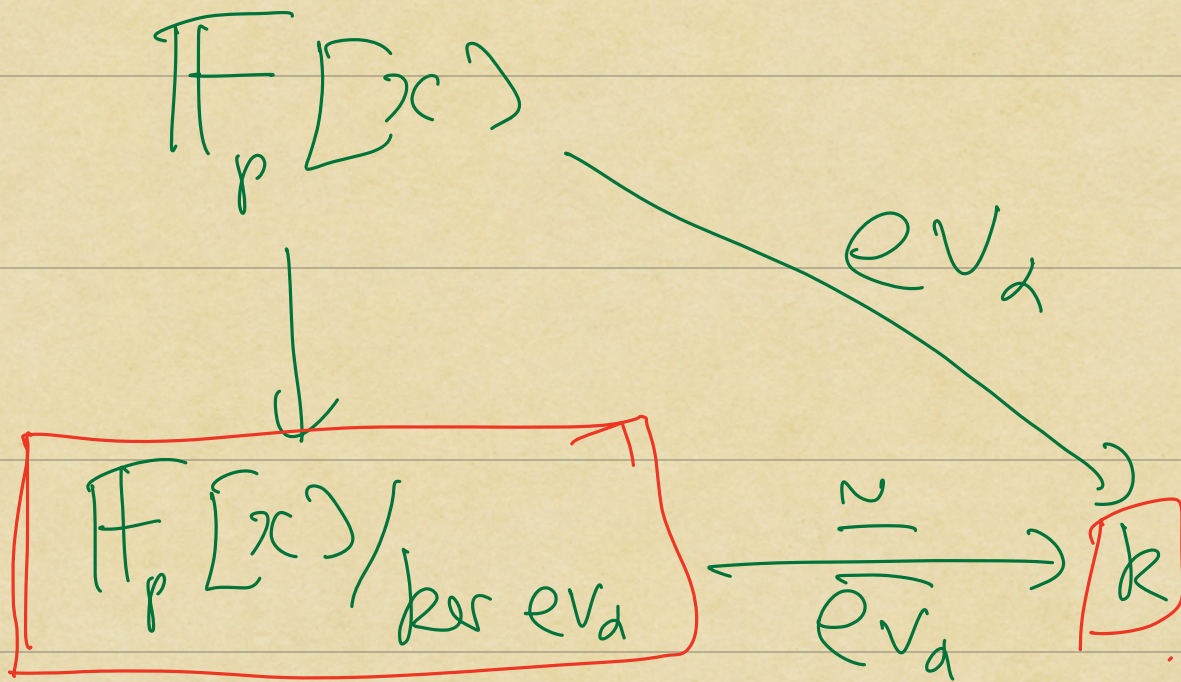
Since $x^d - 1$ is a non-zero polynomial in $k[x]$ of deg d , HW 1 tells us

$$\left| \{ \alpha \in k^\times : \alpha^d = 1 \} \right| \leq d.$$

$$\mathbb{F}_p \leq k \quad \left(\begin{array}{l} \text{characteristic} \\ \text{of } k = p \end{array} \right)$$

Observation (2')

$\alpha \in k^x$
generatur



What can we say about
 $\ker \text{ev}_\alpha$??

Ideals & quotient rings

We know: $\forall f(x) \in R[x]$,

$$\text{and } (f(x)) = \left\{ c(x) f(x) : c(x) \in R[x] \right\}$$

Then $\overbrace{R[x]}^{(f(x))}$ has the structure

of a quotient ring.

Question: R : Commutative ring
 $I \leq R$: subgroup

When is $R \xrightarrow{\pi} R/I$

a homomorphism of rings?

Answer:

We need

$$\pi(r_1, r_2) \stackrel{??}{=} \pi(r_1) \cdot \pi(r_2)$$

$$r_1 r_2 + I \stackrel{??}{=} (r_1 + I) \cdot (r_2 + I)$$

$$r_1 + I = r_1 + j_1 + I$$

$$r_2 + I = r_2 + j_2 + I$$

??
..

$$j_1, j_2 \in I$$

??
..

$$(r_1 + j_1)(r_2 + j_2) + I \stackrel{??}{=} (r_1 + j_1 + I) \cdot (r_2 + j_2 + I)$$

$$r_1 r_2 + \boxed{j_1 r_2 + r_1 j_2 + j_1 j_2} + I$$

$$\boxed{r_1 r_2} + I \stackrel{??}{=} \boxed{r_1 r_2} + \boxed{j_1 r_2 + r_1 j_2 + j_1 j_2} + I$$

So the condition we need is :

$$\begin{aligned} \forall j_1, j_2 \in I \\ r_1, r_2 \in R \end{aligned}$$

$$j_1 r_2 + r_1 j_2 + j_1 j_2 \in I$$

Observation (3)

This condition is equivalent to :

$$\begin{aligned} \forall j \in I, r \in R \\ rj \in I \end{aligned}$$

Defⁿ A subgroup $I \leq R$
is an ideal if $\forall i \in I, r \in R,$
 $ri \in I.$

Fact: $R \longrightarrow R/I$

is a homomorphism of

rings $\Longleftrightarrow I \leq R$
is an ideal

Example: (1) $(f(x)) \leq R[x]$

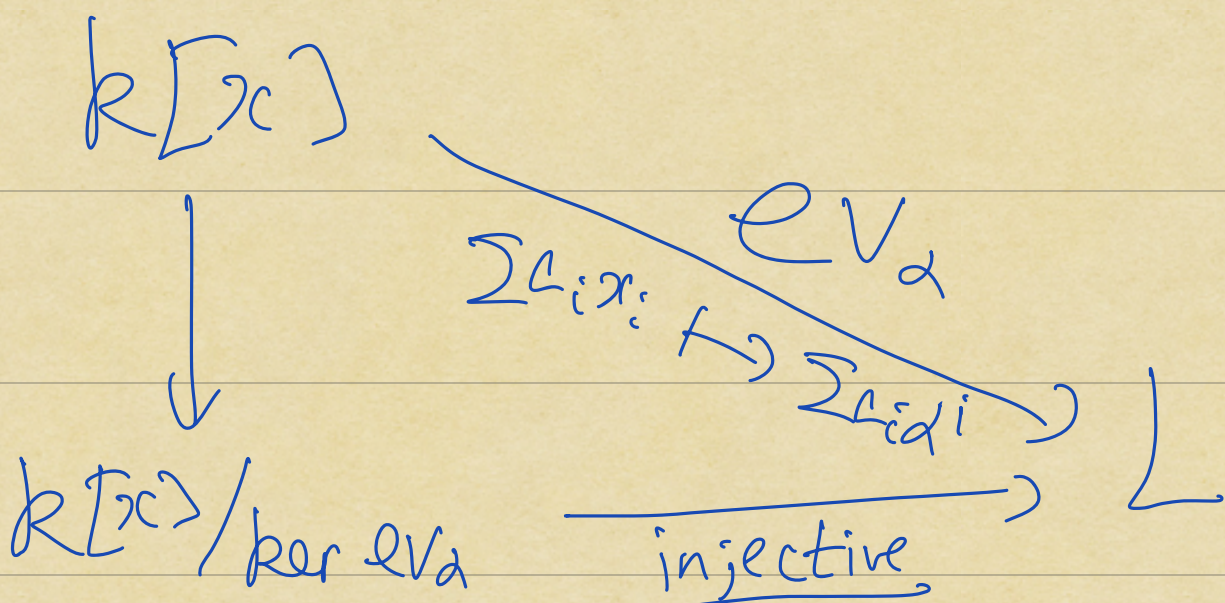
is an ideal in this sense.

$(f(x)) = \{a(x)f(x) : a(x) \in R[x]\}$

(2) $H \leq \mathbb{Z}[x]$: automatically
 $\parallel$
 $d\mathbb{Z}[x]$
 an ideal

$\Rightarrow \mathbb{Z}[x]/d\mathbb{Z}[x]$ is a quotient ring.

(3) $K \leq L$: field extension
 ψ
 α



Then: $\ker \text{ev}_\alpha \leq K[x]$ is an ideal

$\Rightarrow K[x]/\ker \text{ev}_\alpha$ has the structure

of a ring

Fact (2)

$\varphi: R \longrightarrow R'$ ring
homomorphism

Then $\ker \varphi \leq R$ is an ideal.

Pf: $s \in \ker \varphi, r \in R$

$$\begin{aligned}\varphi(rs) &= \varphi(r)\varphi(s) \\ &= \varphi(r) \cdot 0 \\ &= 0\end{aligned}$$

$\Rightarrow rs \in \ker \varphi$